

K-4 Installation Procedures

This section describes what to do when the FastBack® Model 260E-G2 arrives at your facility. The following topics are addressed in this section:

- Installing the Drive Assembly
- Installing the Pan Assembly
- Final Positioning
- Wiring the Utility Enclosure
- Adjusting Gate Operation
- Final Installation Checklist

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Installing the Drive Assembly

The drive assembly is shipped in a crate that consists of five (5) plywood panels and a base pallet.

Tools Needed:

- #2 Phillips bit
- drill
- socket set
- wrench
- forklift

<u>Step</u>	<u>Description</u>
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- | | |
|---|--|
| 1 | Remove top of the crate, using the drill. Loosen the screws from the top of the shipping crate and remove the top (see Photo 4-1). |
|---|--|



Photo 4-1 – Removing the Screws at the Top of the Shipping Crate

Instructions for disassembling the crate may be written inside the crate, as shown in Photo 4-2.



Photo 4-2 – Instructions Inside the Crate

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- 2 Remove the ends and sides of the crate. Using the drill, loosen the screws holding the ends of the crate and remove the ends (see Photo 4-3).



Photo 4-3 – Removing the Ends of the Crate

Loosen the screws holding the sides of the crate, and remove the sides.

- 3 Unbolt the articulating feet from the pallet making up the bottom of the crate.
- 4 Prepare to lift the drive assembly off the pallet. Position the forklift between the articulating feet on the underside of the drive assembly.

Note: *The drive assembly can also be lifted using a forklift and straps.*



CAUTION

If using lifting straps and a forklift, remove the pan first to avoid damaging the pan.

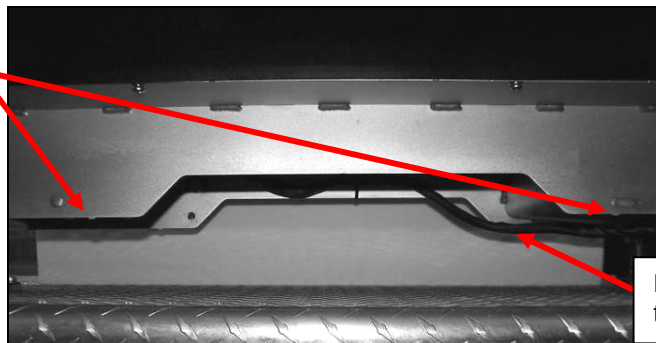
Note: *If your FastBack® arrived with the pan attached to the drive assembly, remove the pan. If the pan was shipped separately, do not attach it until the drive assembly is positioned.*



CAUTION

Make sure that the forks clear the power cord from the utility enclosure to the motor (see Photo 4-4).

Fork-Lift Lift
Points



Power Cord
to Motor

Photo 4-4 – Forklift Lifting Points

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- 5 Move the drive assembly into position. Carefully lift the drive assembly off the pallet and place it in its designated location.
Note: *The drive assembly must be properly oriented front to rear. Refer to the equipment layout drawing for the proper orientation.*
- 6 Attach the drive assembly to the support member. Loosely bolt the drive assembly to the appropriate support member (T-pedestal or angle iron).
- 7 Verify the shipment is complete (if it is the original shipment).

If you are uncrating the original shipment, the drive assembly crate also contains a custom timing plate (part no. 30593041). The timing plate is used when replacing the eccentric belt (see Photo 4-5).

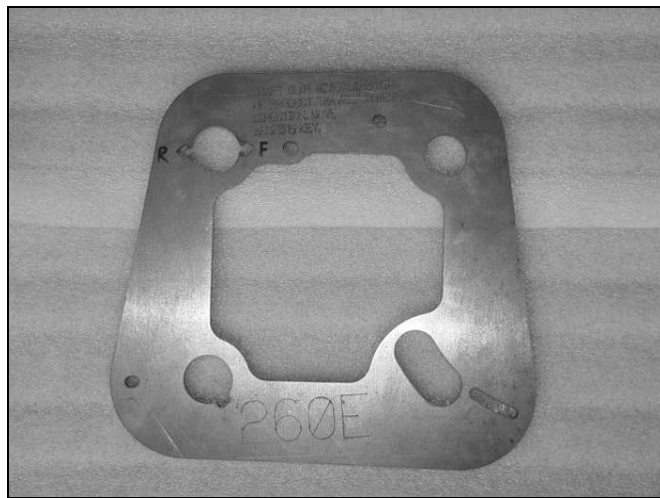


Photo 4-5 – Timing Plate (Part Number 30593041)

Note: *Only one timing plate is included with the original shipment. Heat and Control recommends storing the timing plate with your facility's other tools or spare parts.*

The original shipment also contains accessories for communicating with, diagnosing, and reprogramming the VFD. These accessories are as follows:

Yaskawa VFD

- One Yaskawa hand-held operator with attached communications cable
- One Yaskawa owner's manual

Note: *Only one of each accessory is included with the original shipment. Heat and Control recommends storing these accessories with your facility's other tools or spare parts.*

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Photo 4-6 shows various accessories as they may have been packed in the crate.



Photo 4-6 – Various Accessories in Crate

If any of these items are missing, contact your project manager or the Heat and Control Galesburg, IL office at (309) 343-5518.

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Installing the Pan Assembly

Note: *Do not install the pan until the drive assembly has been moved into position as described in “Installing the Drive Assembly.”*

Tools Needed:

- torque wrench
- socket wrench
- 11/16-inch [17 mm] socket
- 3/4-inch [19 mm] socket
- 11/16-inch [17 mm] open ended wrench

Step	Description
1	Remove the lid of the pan assembly shipping crate. It is recommended that no fewer than two people, carefully lift the pan from the crate.
2	Loosen doubler plate bolts. At each arm connection point, use an 11/16-inch [17 mm] socket to loosen the doubler plate bolts to allow room for the pan to be lowered into position around the upper arm bushings.
3	Remove the pan attaching bolt using a 3/4-inch [19 mm] socket.
4	Place the pan brackets onto the drive arm of the drive unit and the opposite brackets located on the side of the pan on the idler support arms. Insert the pan attaching bolts through doubler plates and arm bushings, tighten them finger tight.
5	Tighten doubler plate bolts, using an 11/16-inch [17 mm] socket. Tighten the two (2) doubler plate bolts.
6	Tighten the pan attaching bolts. Beginning with the drive connecting point, use a 3/4-inch [19 mm] socket with the torque wrench set at 42 ft-lbs (57 N-m) to tighten pan attaching bolts (see Photo 4-7).



CAUTION

The pan may be damaged if it tips before all attachment points are tightened.

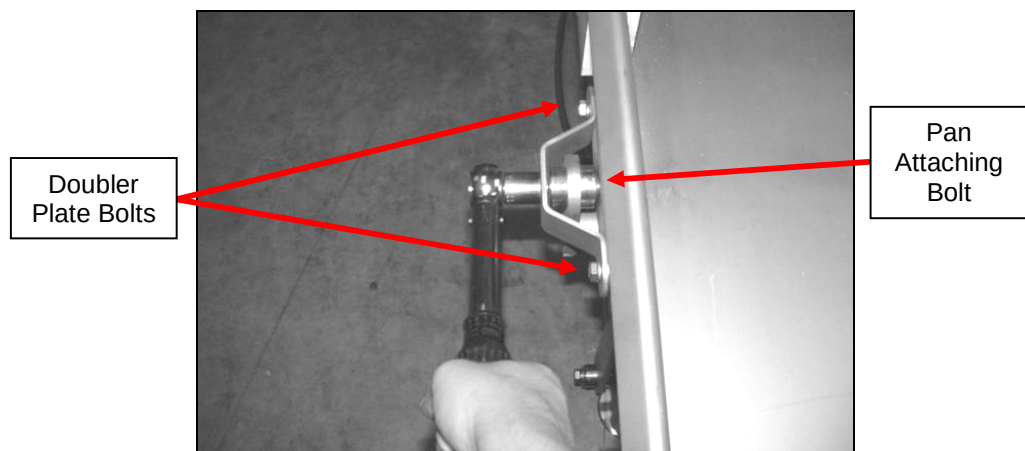


Photo 4-7 – Tightening the pan attachment points

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- 7 Reinstall adjustable nose discharge if necessary

The adjustable nose discharge, if included, may have been reversed for shipping (see Photo 4-8).



Photo 4-8 – Shipping Position of Adjustable Nose Discharge

Reinstall the adjustable nose discharge as follows:

- a. Fully loosen the four-prong knobs on each side of the pan.
- b. Slide the adjustable nose off the pan.
- c. Turn the nose around and slide it back onto the pan. Make sure that the nose is between the two white washers on the four-prong knob (see Photo 4-9).



Photo 4-9 – Correctly Installed Adjustable Nose Discharge

- d. Retighten the four-prong knobs.

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Final Positioning

This procedure describes how to perform a final check of the FastBack® Model 260E-G2's position.

Note: *The equipment layout drawing for your facility's specific installation supersedes the general guidelines below.*

Step	Description
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- | | |
|---|--|
| 1 | Locate a copy of Heat and Control's Equipment Layout Drawing (ELD). This is the master dimensional guide for positioning the FastBack® conveyors. |
| 2 | Verify that the center line of the drive and pan assemblies lines up with the specified location in the plan view of the Equipment Layout Drawing. |
| 3 | Verify that the rear of the pan is positioned so that it catches the product from the in-feeding equipment. Verify the discharge of the pan adequately overlaps the next piece of equipment to prevent spillage. |
| 4 | Align the pan and drive in side view, according to the dimensions on the equipment layout drawing. |

If the idler arm is on-board the drive unit, then adjusting the drive feet up/down will control the pan height.

If the idler arm is remote (not on the drive), then use the adjustable foot on the remote idler to position the pan height (see Photo 4-10), in addition to adjusting the drive unit feet up/down.

The angle of the pan is usually specified in the ELD to be less than a 1.5° incline. The closer the pan is to level, the faster the product will travel on it.

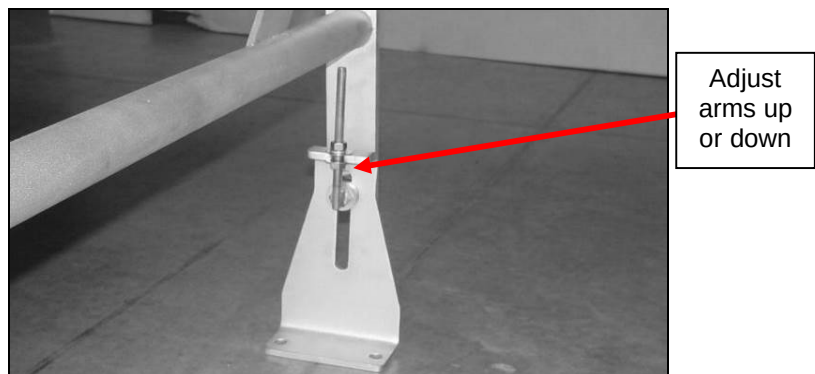


Photo 4-10 – Adjustable foot found on remote idler arm

Note: *Inclining the pan more than 1.5° may impede travel rates.*

- | | |
|---|--|
| 5 | Ensure the top of the drive unit is parallel to the pan. Adjust the three (3) articulating feet as required. |
| 6 | Tighten the adjustable feet on the support legs. After verifying that the FastBack® is properly positioned, tighten the jam nuts on the adjustable feet. Use a wrench to hold the threaded stem from turning while tightening the jam nut. |

Wiring the Utility Enclosure

This procedure describes how to connect power and control wiring to the on-board utility enclosure, which typically houses the following:

- Yaskawa V1000 VFD
- Up to two (2) Line Reactors (480 V applications ONLY when the VFD is remotely mounted)
- When applicable, Pneumatic pressure regulator and solenoid valves
- When applicable, Ethernet communication cables
- Terminal strip for control wiring, including VFD controls, discreet photoeyes, ultrasonic sensors (analog output), and position transducers.

The Yaskawa V1000 comes from the factory programmed to run at 60 Hz in single-speed, start/stop mode with a run confirmation signal.

Tools Needed:

- #2 Phillips head screwdriver

Step	Description
------	-------------

- | | |
|---|--|
| 1 | Open the utility enclosure, by removing the eight screws holding the lid of the utility enclosure in place (see Photo 4-11). |
|---|--|



Photo 4-11 – Removing lid of utility enclosure

- | | |
|---|--|
| 2 | Locate the electrical schematic drawings. Use the schematic drawings supplied for this project to determine the scope of the installation and other specifics (voltage, etc.). |
|---|--|

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3 Install the fittings (as required).

The bottom of the utility enclosure box contains multiple holes to accommodate the full spectrum of electrical connections that may be required for a particular FastBack® conveyor. See Figure 4-1 for details. Identify which holes are to be used for this conveyor and plug all unused holes. The utility panel is supplied with fittings in the holes as shown in the figure below. The only fitting that needs to be supplied in the field for a typical installation is for the main power infeed to the disconnect switch(es).

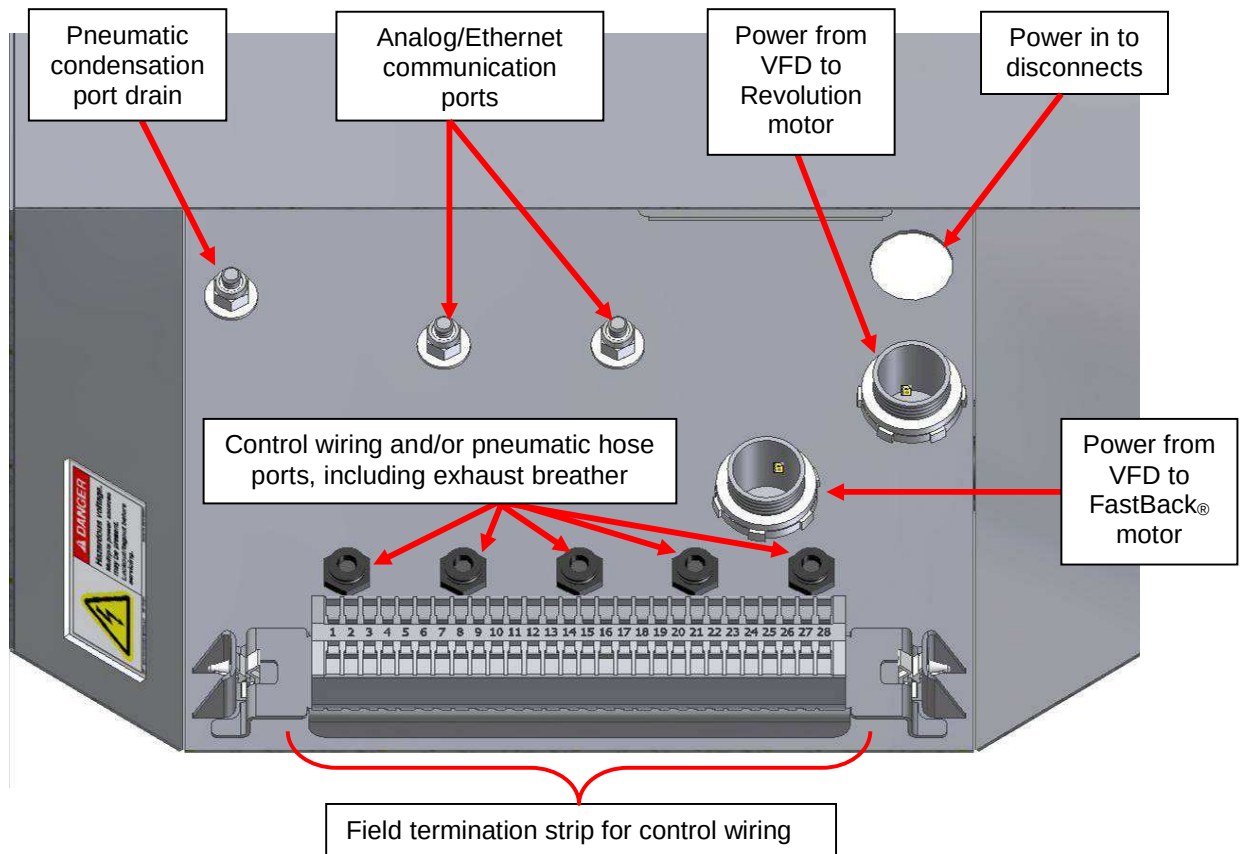


Figure 4-1 – Bottom of Utility Enclosure Designating Power, Control, and Pneumatic Ports

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4 Connect primary power leads.

When the FastBack® is supplied with a VFD installed “On-board” in the utility enclosure, the unit comes pre-wired with the following 3-phase power connections (see Figure 4-2):

- From disconnect switch to VFD input terminals
- From VFD output to motor

Note: *The Incoming 3-phase power to the disconnect switch is the primary field wiring to install.*

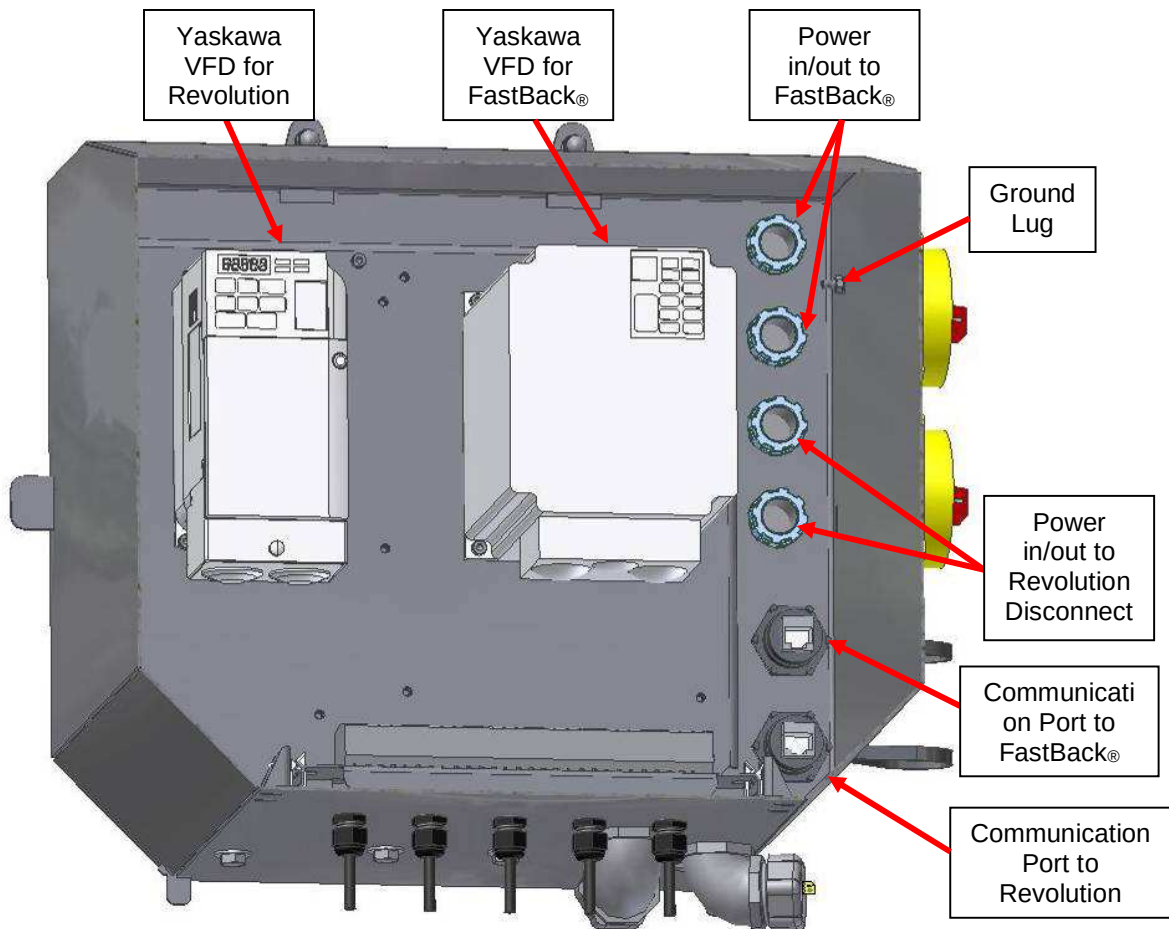


Figure 4-2 – Interface fittings to disconnect switches and communication ports for a utility enclosure with two (2) Yaskawa V1000 VFD's

- Connect the primary power leads to terminals 1, 3, and 5 of the red and yellow disconnect switch. If a Revolution unit is integrated into the FastBack® conveyor, then a 2nd motor and disconnect switch will be supplied within the utility enclosure.
- Connect the incoming ground wire to the grounding lug on the inside wall of the utility enclosure box.

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5 Connect VFD control leads

Connect the control leads as described in Table 4-1 (for Yaskawa VFD). Wiring diagrams for the Yaskawa V1000 inverter appears in TechPub PH009.

Application	Terminal/Connection
Start/stop All applications (2 wires)	Install a dry contact closure (relay or switch) between terminals SC (digital common) and S1 (run forward).
Single speed Speed select (no wires needed)	No additional wiring needed; conveyor will run at speed set in parameter n024 (60 Hz).
Multi-speed Preset speed select (2 wires)	Install a pair of wires to provide dry contact closure (relay or switch) between terminals SC and S5 (preset speed reference 1) and/or SC and S6 (preset speed reference 2). Refer to the preset speed table in the Yaskawa wiring diagram (TechPub PH009) to determine the applicable binary combination.
Variable speed 4-20mA analog input (shielded/twisted pair)	Connect leads (must be a shielded/twisted pair) to AC (analog common) and A2 (analog frequency reference input in 4-20 mA). A hardware and parameter change is needed to complete the installation. <u>Step 5</u> of this procedure describes how to make the needed changes.
Run confirmation All applications, optional (2 wires)	Connect two leads to MA and MC. This is a relay output, which closes when the inverter output is on.

Table 4-1 – Wiring Instructions for Yaskawa V1000 VFD

6 Reconfigure the V1000 inverter for variable speed if necessary (optional).

For those FastBack® Model 260E-G2 applications requiring variable speed controls, the inverter must be reconfigured. The following procedure will guide you through the necessary adjustments that will enable the inverter to respond proportionately to a 4-20mA analog speed reference input.

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Yaskawa V1000 – Variable Speed (4-20mA analog input) Application

- a. Open the utility enclosure.
- b. Set the S1 DIP switch to “I” (right side), as shown in Photo 4-12.

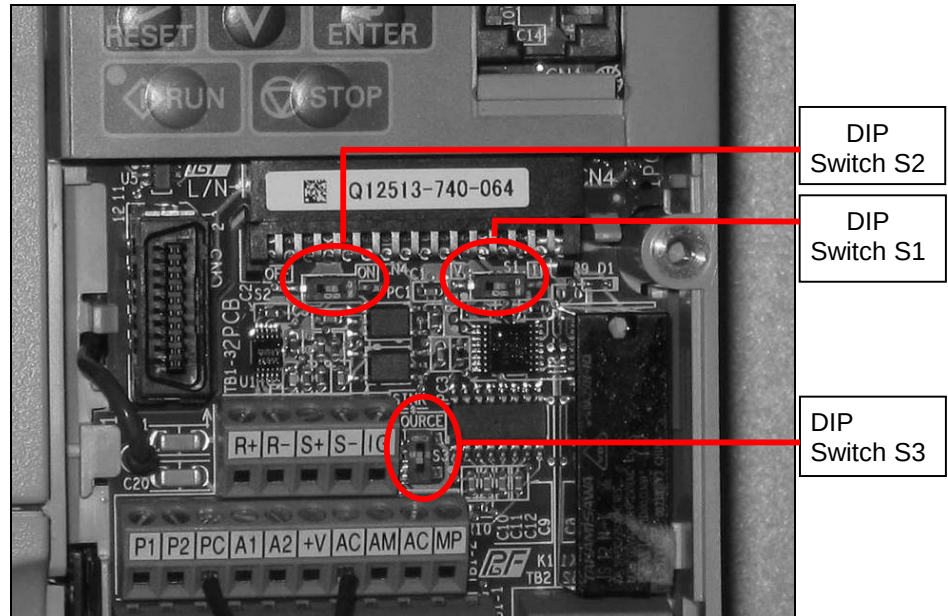


Photo 4-12 – Setting DIP Switches for Variable Speed Control

- c. Set the S2 DIP switch to ON (right side), as shown in Photo 4-12.
- d. Connect 4-20 mA analog input wires to terminals A2 and AC.
- e. Connect power to the FastBack®.
- f. Per the parameter values in **Technical Publication PH009**, change parameter B1-01 from “0” to “1”.

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Alternate VFD placement is shown below.

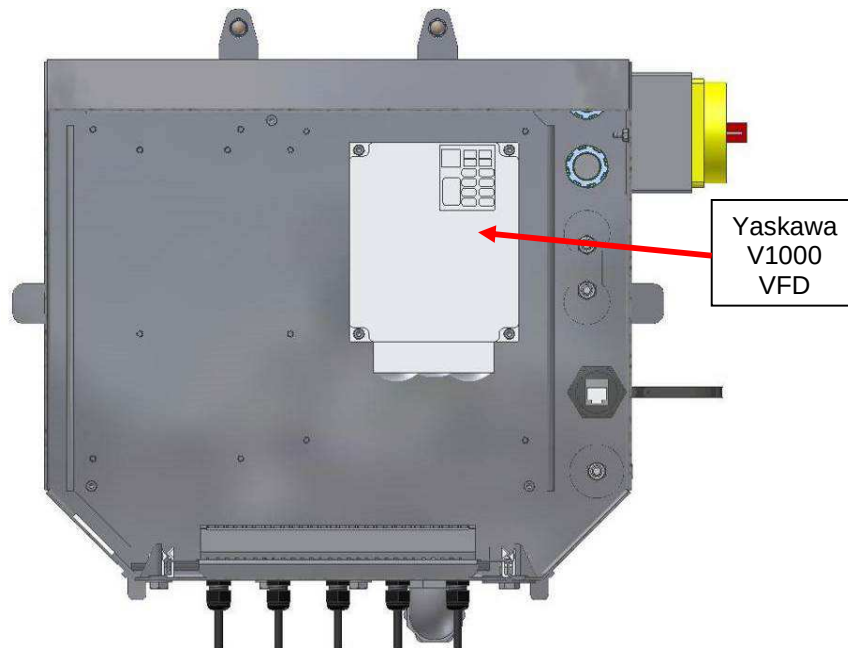


Figure 4-3 – Utility Enclosure with a Single Yaskawa V1000 VFD for the FastBack® Drive

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In applications where the VFD's are NOT mounted on-board the FastBack® drive AND 480V power is being used, the utility enclosure will house a Line reactor unit for each motor that is part of the FastBack®. The purpose of these line reactors is to prevent extremely high voltage spikes from hitting the motor and possibly damaging it. The line reactors are required to be installed in these applications to maintain the Heat and Control, Inc. warranty.

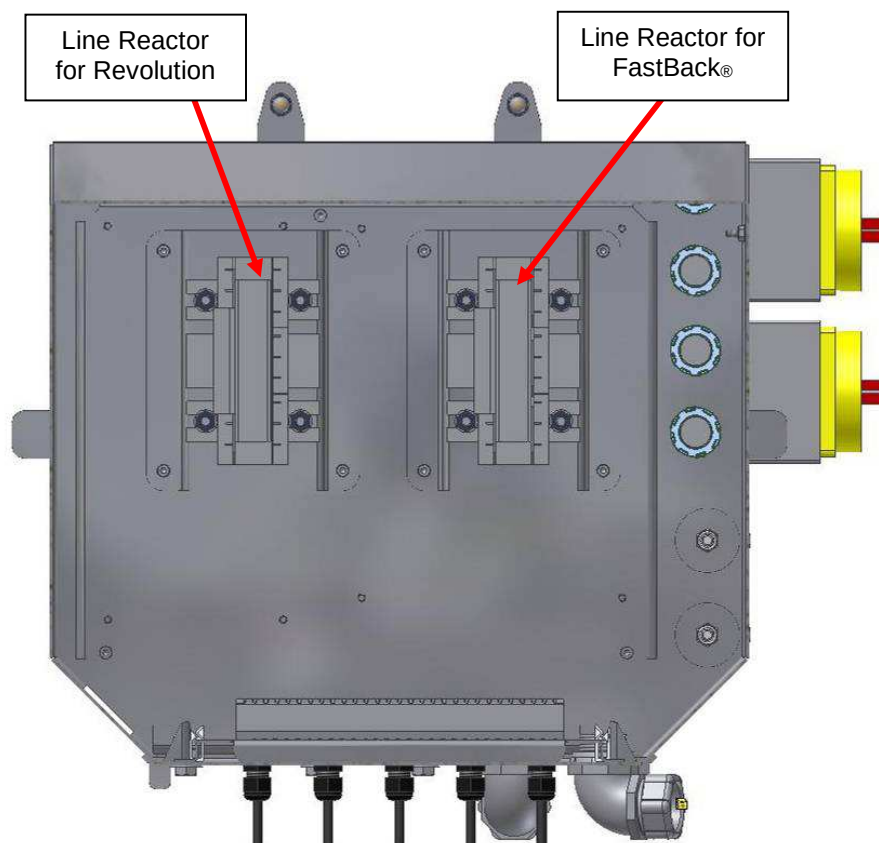


Figure 4-4 – Utility Enclosure with two (2) Line Reactors (ONLY for 480 V Applications where the VFDs are Remotely Mounted)

- 7 If there are no other control wires to terminate, replace the lid of the utility enclosure box by fastening the eight (8) screws. If there are sensors and/or gates on this conveyor, leave enclosure open – See next section.
- 8 See **Technical Publication PH009** for other set-up procedures and parameter settings for the Yaskawa V1000 VFD.

See **Technical Publication PH015** for other set-up procedures and parameter settings for the Allen-Bradley PowerFlex 525 VFD.

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Adjusting Gate Operation

This procedure describes how to connect the gate and gate controls to all sources at your facility and make all necessary adjustments for operation.

Tools Needed:

- small standard screwdriver
- #2 Phillips head screwdriver

<u>Step</u>	<u>Description</u>
-------------	--------------------

- | | |
|---|--|
| 1 | With the pan in place, find the infeed end of the air lines (factory-installed along the pan) and connect the air fittings located on the drive unit in sequence to the gate fittings by snapping the quick disconnect in until it clicks (see Photo 4-13). Ensure that there is ample slack in the air lines between the stationary drive and the moving pan to prevent a tugging on the lines that would cause premature wear. |
|---|--|

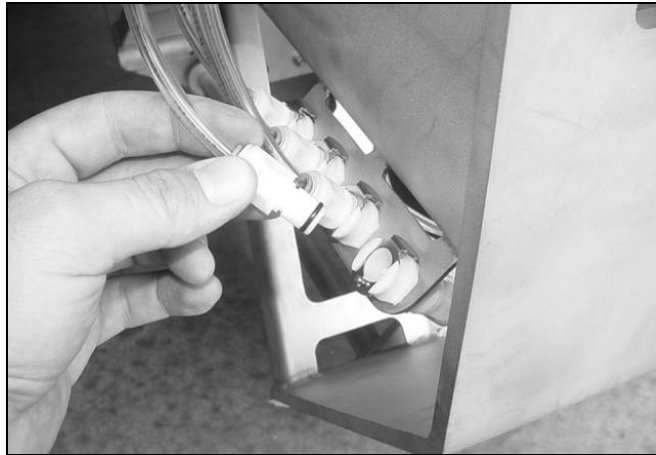


Photo 4-13 – Air fitting quick disconnect

- | | |
|---|--|
| 2 | Open the utility enclosure per the previous section. |
|---|--|

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- 3 Connect air supply to the unit. Route supply air tubing (1/4 inch or 6mm) up through one of the cord grips supplied with the utility enclosure and connect to the pressure regulator (see Figure 4-5).

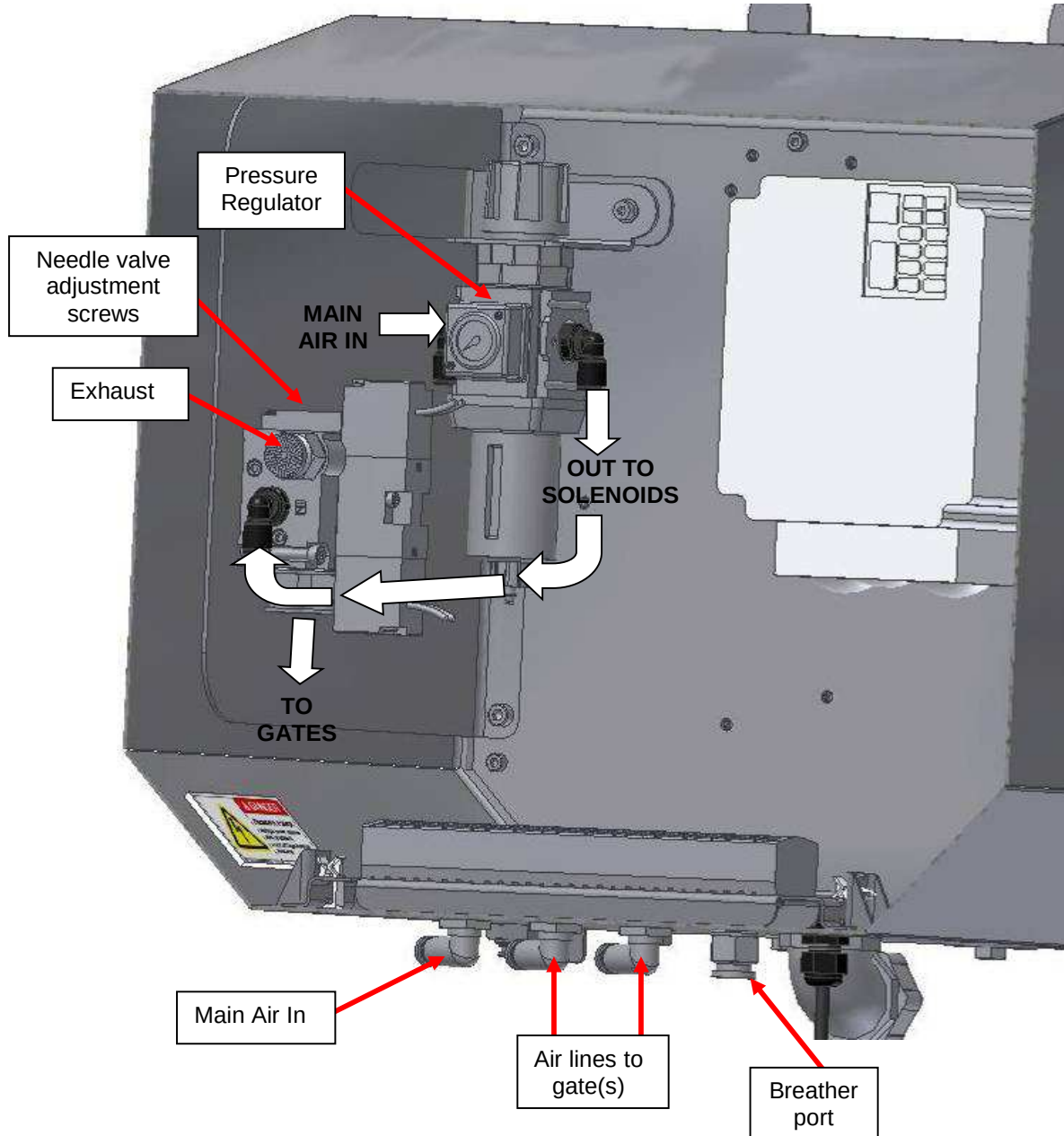


Figure 4-5 – Utility Enclosure with Typical Pneumatic Package

- 4 Regulate system pressure to 30 PSI (2.1 kgf/cm²).
Set the system supply pressure to a pressure of 30 PSI (2.1 kgf/cm²) using the pressure control knob. The pressure will read out on the pressure regulator gauge (see Figure 4-5).

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- 5 Confirm air lines to the gates are secure. The air lines from the gates to the utility enclosure should be factory-installed, but confirm that these lines are in place and secure to the quick-connect fittings supplied in the bottom of the enclosure.
- 6 Wire the solenoid control wires. Using the electrical schematic drawings and the FastBack® Top Level Assembly (TLA) drawing, wire the field control wires to the terminal strip inside the utility enclosure.
- 7 Make adjustments to manifold screws. Using a small standard screwdriver, make the appropriate pressure adjustments for gate control by turning the flow control needles on the solenoid junction, adjusting the air pressure to slightly more than what is needed to hold the gate forward when it is closed. Turning the pressure down this low will necessitate an adjustment to the manifold flow control screws, so that the gate closes in a timely manner. Photo 4-14 shows where the screws are located on the manifold. One screw controls the gate opening speed while the other screw controls the gate closing speed. Both flow control screws adjust the exhaust rate from the unpressurized side of the air cylinder. Using a small standard screwdriver, you should always start with the screws lightly seated and then back off (counter-clockwise) approximately 3 turns open. Adjust from that point as needed.

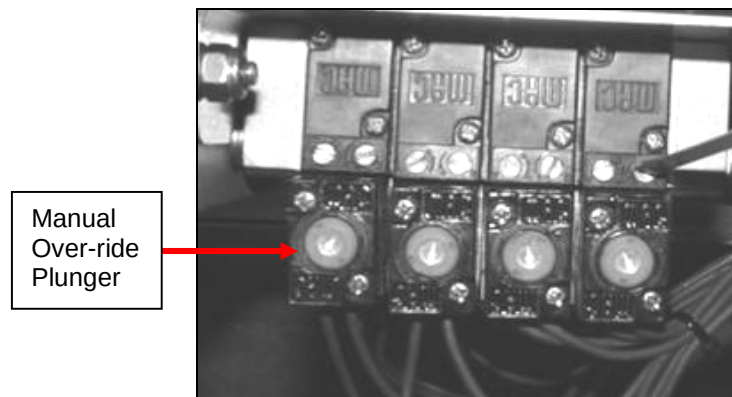


Photo 4-14 – Adjustment of manifold screws

- 8 Cycle the gates using manual over-ride plunger. To test the gates operation with the flow control valves adjusted, press in the over-ride plunger on the solenoid (see Photo 4-14). Adjust the air pressure and/or needle valves as needed to get optimum gate cycling.
- 9 If necessary, make mechanical gate adjustments. Adjust the gates end of stroke position with the clevis on the end of the cylinder rod so that at the end of the closing stroke the ridge on the UHMW gate slide is just touching the edge of the pan cutout. Then, adjust the cylinder rod nut 1/4-turn further in the adjustment. This will ensure a snug fit between the gate and the pan cutout.

If the gate is sluggish or not sliding smoothly, loosen the rail support bolts and the bushing block bolts. Allow the rails and gate tray to float to their position of least resistance. Tighten the bolts one at a time while stroking the gate open / closed. Once all the bolts are fastened tight, ensure the gate opens and closes smoothly. Repeat this process as needed.

Final Installation Checklist

- ☐ All conduits between platforms are flexible.
- ☐ Support platforms are in accordance with equipment layout drawings.
- ☐ Cross-feeders are centered on dispersion tables and are at the elevation specified in equipment layout drawings.
- ☐ Distribution conveyor gates are clear of cross-feeder sidewalls.
- ☐ Distribution conveyors are perpendicular to cross-feeders whenever possible.
- ☐ Maximum vertical gap between interfacing distribution conveyors is 1/4-inch.
- ☐ Photoeyes are aimed at the apparent apex of product discharge without being obstructed by the waterfall effect of product.
- ☐ Enclosures are sealed for watertightness. Wires are clear of seals.
- ☐ Rub points or collision points have been eliminated.
- ☐ Are all 260E-G2 pan bolts tight?
- ☐ Are pan grounding straps installed?
- ☐ Is drive positioned per ELD dimensions?
 - ☐ Level pan side-to-side
 - ☐ Center pan side-to-side
 - ☐ Position pan front-to-back
 - ☐ Angle pan front-to-back as required
 - ☐ Confirm / measure final position
- ☐ Is there near-zero foot bounce when running? (If not, adjust feet.)
- ☐ Are remote idler arms at right angle to pan in neutral position?
- ☐ Are adjustable feet secured down? Are jam nuts tightened?
- ☐ Will FastBack® run in manual from OIT?
- ☐ Is motor rotation CCW? (If not, reverse leads.)
- ☐ Does “something” travel the correct direction on the pan?
- ☐ For this voltage, are motor rated amps in VFD correct? (Variable E2-01)
- ☐ Are all control wires terminated properly?
- ☐ Does disconnect switch turn off unit?
- ☐ Are all utility enclosure lid screws tightened?